

2501-PYL1003 (Mechanics & Special Theory of Relativity)

Practice Mid-Semester Examination — Paper 1 (Unofficial, Instructor-Style)

Full Marks: 60 Time: 2 hours Set: Practice-1

Instructions

An MCQ can also be an MSQ (select all correct options). Justify every choice — no marks for a bare letter. Show all intermediate steps and reasoning.

State any approximation or rounding explicitly, with justification.

Numerical answers should be rounded to two decimal places unless stated otherwise.

****1.**** A physical quantity Ξ is defined through the relation

$$\Xi = \frac{\hbar^a G^b c^d}{k_B}$$

where $\hbar$ is the reduced Planck constant, G the gravitational constant, c the speed of light, and k_B the Boltzmann constant. If Ξ is required to have the dimension of **temperature**, determine the exponents a, b, d using dimensional analysis. ***(No numerical value of $\hbar, G, c, k_B$ is needed — work purely with M, L, T, and thermodynamic temperature Θ as fundamental dimensions, noting $[k_B] = \text{ML}^2\text{T}^{-2}\Theta^{-1}$.)** [5]****

2 (MSQ). Consider a quantity Y with dimension $\text{M}^1\text{L}^{-1}\text{T}^{-2}$ (pressure/stress). Which of the following combinations is/are dimensionally consistent with Y ? Justify each. a) ρv^2 , where ρ is mass density and v is speed b) F/L^2 , where F is force and L is length c) $\eta\omega$, where η is dynamic viscosity ($\text{ML}^{-1}\text{T}^{-1}$) and ω is angular frequency d) $Y_{\text{elastic}} \cdot \theta$, where Y_{elastic} is Young's modulus and θ is a dimensionless strain **[4]**

3. A particle of mass 0.25 kg moves along a straight line. Starting from rest at $t = 0$, it is subjected to a **jerk** (rate of change of acceleration) that varies as $j(t) = 6 - 4t \text{ m/sec}^3$, valid until the acceleration first returns to zero at some time $t = t_1 > 0$. Given that the acceleration at $t = 0$ is $a(0) = 0$: (i) Find t_1 . (ii) Find the velocity and displacement of the particle at $t = t_1$. (iii) Determine whether the particle is speeding up or slowing down at $t = t_1^+$ (just after t_1), with justification. **[2+2+2 = 6]**

4. Two blocks, A (mass 2 kg) and B (mass 8 kg), initially at rest, are pushed by *the same time-dependent force* $F(t) = 5t \text{ N}$ applied for $t \in [0, 4] \text{ sec}$ along a frictionless horizontal floor. (i) Find the ratio of distances travelled by A to B during this interval. (ii) At $t = 4 \text{ sec}$, the force is removed. Find the ratio of kinetic energies $KE_A : KE_B$ at that instant. **[3+2 = 5]**

5 (MSQ). A point Q has Cartesian coordinates $(x, y, z) = (-6, 8, -\sqrt{300})$. Let (ρ, ϕ, z) be its cylindrical polar coordinates and (r, θ, ϕ) its spherical polar coordinates (with $z = r \cos \theta$, ϕ measured from the $+x$ -axis toward $+y$). Which statement(s) below is/are correct (angles rounded to two decimals)? a) $\rho = 10$, $\phi = 126.87^\circ$ b) $r = 20$, $\theta = 150.00^\circ$ c) $r = 20$, $\theta = 120.00^\circ$ d) $\phi = 233.13^\circ$ **[5]**

6. A vector field is given in spherical polar coordinates as $\vec{A} = r \hat{r} - r \sin \theta \hat{\theta} + 4 \hat{\phi}$. (i) Derive general expressions for $\hat{r}$, $\hat{\theta}$, $\hat{\phi}$ in terms of $\hat{i}$, $\hat{j}$, $\hat{k}$ (you may state the standard result, but show the differentiation $\hat{u} = (\partial \vec{r} / \partial u) / |\partial \vec{r} / \partial u|$ for **one** of the three explicitly). (ii) Evaluate $\vec{A} \cdot \hat{k}$ (a scalar) at the point $r = 2$, $\theta = 60^\circ$, $\phi = 0^\circ$. (iii) Express the full vector $\vec{A}$ at that point in Cartesian components (A_x, A_y, A_z) . **[3+2+3 = 8]**

7. A comet orbits the Sun on a bound elliptical trajectory. At two points on its orbit where the radius vector makes angles $\theta_1 = 0$ and $\theta_2 = \pi/2$ with the major axis (measured from perihelion), the comet's *radial* distances are found to satisfy $r(\pi/2) = 3 r(0)$. Using $r(\theta) = p / (1 + \epsilon \cos \theta)$, determine the eccentricity ϵ of the orbit. Show your algebra fully; do not merely quote the apsidal formula. **[5]**

8. For a general central force orbit $r(\theta) = p / (1 + \epsilon \cos \theta)$ with $\epsilon < 1$: (i) Prove, starting from conservation of angular momentum $L = m r^2 \dot{\theta}$, that the areal velocity dA/dt is constant. (ii) Hence derive an expression for the orbital period T in terms of p , ϵ , L , and m (semi-major axis $a = p / (1 - \epsilon^2)$ may be used without re-derivation). **[3+4 = 7]**

9. A ball is dropped from rest from a tower of height $h = 100$ m at a location with geographic latitude $\lambda = 30^\circ$ N on Earth (angular velocity of Earth's rotation $\Omega = 7.29 \times 10^{-5}$ rad/sec, $g = 9.8$ m/sec²). (i) Write down Newton's second law in the rotating Earth frame, identifying the Coriolis term explicitly. (ii) Using the leading-order (zeroth-order-in- Ω) trajectory to evaluate the Coriolis correction, find the deflection of the ball from the vertical (magnitude and compass direction) when it lands. Neglect centrifugal effects. **[3+5 = 8]**

10. A small bead of mass m is constrained to slide without friction along a rigid, straight rod that lies in a horizontal plane and rotates about a fixed vertical axis through one end at a **constant** angular velocity Ω . At $t = 0$, the bead is at radial distance r_0 from the axis with zero radial velocity. (i) Working in the rotating frame attached to the rod, write down the radial equation of motion for the bead, identifying the centrifugal term. (ii) Solve for $r(t)$. (iii) Find the magnitude of the Coriolis force on the bead as a function of time (you need not solve for the constraint force fully — express it in terms of $\dot{r}(t)$). **[2+4+2 = 8]**

11. A rocket of initial total mass M_0 (structure + fuel) starts from rest in free space (no gravity, no drag) and ejects mass at a **constant rate** $\mu = -dm/dt$ with exhaust speed **relative to the rocket** u , constant. Using the variable-mass form of Newton's second law: (i) Derive the Tsiolkovsky rocket equation for the rocket's speed $v(t)$ in terms of u , M_0 , and instantaneous mass $m(t)$. (ii) If the rocket burns fuel until only 20% of M_0 remains, and $u = 2500$ m/sec, find the final speed. **[4+2 = 6]**

12. A uniform flexible chain of total length L and mass per unit length λ is piled up on a table, with one end raised and pulled vertically upward at a **constant speed** v_0 by an external agent (links already off the table are perfectly straight and moving at v_0 ; links still on the table are at rest and are picked up instantaneously as the moving end passes them). Let $x(t)$ be the length of chain already lifted off the table, with $x(0) = 0$. (i) Using the variable-mass equation of motion (accounting for the momentum carried by newly-added

mass), find the force $F(x)$ that the agent must apply as a function of x , in terms of λ , v_0 , x , and g . (ii) Evaluate F at the instant $x = L/2$ for $\lambda = 0.5 \text{ kg/m}$, $v_0 = 2 \text{ m/sec}$, $g = 10 \text{ m/sec}^2$. **[5+2 = 7]**

13. In the pulley system shown conceptually below, pulley 1 is fixed to the ceiling; pulleys 2, 3, and 4 are **movable** and arranged so that the cable from pulley n wraps around pulley $n+1$ before terminating (pulley 4's cable end applies the 10 N pull, as in the reference exam's Q8, but here there is a **4th** movable pulley added below). All pulleys and cables are massless and frictionless; block B hangs from pulley 4. (i) Derive a general formula for the mechanical advantage (ratio of block weight to applied force) of such an n -movable-pulley cascade (i.e., 1 fixed + n movable, cable re-routed through each successive pulley), and verify it for $n = 3$ against the given 10 N force system. (ii) Hence compute the mass M of block B and the tension in the topmost cable, given the applied force is 10 N and $g = 10 \text{ m/sec}^2$. **[4+3 = 7]**

14 (MSQ). A block rests on a rough horizontal table ($\mu_s = 0.4$, $\mu_k = 0.3$) and is connected over a frictionless, massless pulley at the table's edge to a hanging block, as in a standard Atwood-with-friction setup. The hanging block has mass 3 kg; the block on the table has mass 5 kg. Which of the following statement(s) is/are correct? a) The system remains static because $\mu_s \times 5 > 3$ (in appropriate units with g cancelling) b) The system accelerates, and the magnitude of acceleration is approximately 0.74 m/sec^2 c) The tension in the string while moving is less than the weight of the hanging block d) If μ_k were exactly 0, the acceleration would be exactly $3g/8$ **[6]**

15. A particle of mass m moves under an attractive central force $F(r) = -k/r^3$ (an inverse-cube force), with angular momentum $L \neq 0$ about the force center. (i) Write down the effective potential $V_{\text{eff}}(r)$ combining the centrifugal term and the potential associated with $F(r)$. (ii) Show that, depending on the sign of $\left(\frac{L^2}{m} - k\right)$, the qualitative character of the effective potential changes fundamentally (i.e., whether a stable circular orbit / turning point can exist at all). State clearly, with reasoning, the condition on k (in terms of L and m) required for a bound turning point to exist. **[3+4 = 7]**

End of Paper 1. Total: 60 marks.